

Fibonacci numbers

Fibonacci numbers

1, 1, 2, 3, 5, 8, 13, 21, 34, 55, . . .

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1							

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1	2						

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1	2	3					

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1	2	3	5				

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1	2	3	5	8			

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib									

```
fib[0] = 1;
```

```
fib[1] = 1;
```

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1								

```
fib[0] = 1;
```

```
fib[1] = 1;
```

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1							

```
fib[0] = 1;
```

```
fib[1] = 1;
```

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1	2						

`fib[0] = 1;`

`fib[1] = 1;`

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1	2						

```
fib[0] = 1;
```

```
fib[1] = 1;
```

```
fib[2] = fib[0] + fib[1];
```

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1	2	3					

```
fib[0] = 1;
```

```
fib[1] = 1;
```

```
fib[2] = fib[0] + fib[1];
```

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1	2	3					

```
fib[0] = 1;
```

```
fib[1] = 1;
```

```
fib[2] = fib[0] + fib[1];
```

```
fib[3] = fib[1] + fib[2];
```

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1	2	3	5				

```
fib[0] = 1;
```

```
fib[1] = 1;
```

```
fib[2] = fib[0] + fib[1];
```

```
fib[3] = fib[1] + fib[2];
```

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1	2	3	5				

```
fib[0] = 1;
```

```
fib[1] = 1;
```

```
fib[2] = fib[0] + fib[1];
```

```
fib[3] = fib[1] + fib[2];
```

```
fib[4] = fib[2] + fib[3];
```

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1	2	3	5	8			

```
fib[0] = 1;
```

```
fib[1] = 1;
```

```
fib[2] = fib[0] + fib[1];
```

```
fib[3] = fib[1] + fib[2];
```

```
fib[4] = fib[2] + fib[3];
```

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1	2	3	5	8			

```
fib[0] = 1;
```

```
fib[1] = 1;
```

```
fib[2] = fib[0] + fib[1];
```

```
fib[3] = fib[1] + fib[2];
```

```
fib[4] = fib[2] + fib[3];
```

```
fib[5] = fib[3] + fib[4];
```

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1	2	3	5	8			

`fib[0] = 1;`

`fib[1] = 1;`

`fib[2] = fib[0] + fib[1];`

`fib[3] = fib[1] + fib[2];`

`fib[4] = fib[2] + fib[3];`

`fib[5] = fib[3] + fib[4];`

`⋮`

`⋮`

`⋮`

Fibonacci numbers

```
fib[0] = 1;  
fib[1] = 1;  
fib[2] = fib[0] + fib[1];  
fib[3] = fib[1] + fib[2];  
fib[4] = fib[2] + fib[3];  
fib[5] = fib[3] + fib[4];  
  ⋮           ⋮           ⋮
```

```
fib[0] = 1;  
fib[1] = 1;  
for( int i = ; ; )  
    ;
```

Fibonacci numbers

```
fib[0] = 1;  
fib[1] = 1;  
fib[2] = fib[0] + fib[1];  
fib[3] = fib[1] + fib[2];  
fib[4] = fib[2] + fib[3];  
fib[5] = fib[3] + fib[4];  
  ⋮           ⋮           ⋮  
  ⋮           ⋮           ⋮  
  ⋮           ⋮           ⋮
```

```
fib[0] = 1;  
fib[1] = 1;  
for( int i = ; ; )  
    fib[i] = + ;
```

Fibonacci numbers

```
fib[0] = 1;  
fib[1] = 1;  
fib[2] = fib[0] + fib[1];  
fib[3] = fib[1] + fib[2];  
fib[4] = fib[2] + fib[3];  
fib[5] = fib[3] + fib[4];  
  ⋮           ⋮           ⋮
```

```
fib[0] = 1;  
fib[1] = 1;  
for( int i = ; ; )  
    fib[i] = fib[i-2] + fib[i-1];
```

Fibonacci numbers

```
fib[0] = 1;  
fib[1] = 1;  
fib[2] = fib[0] + fib[1];  
fib[3] = fib[1] + fib[2];  
fib[4] = fib[2] + fib[3];  
fib[5] = fib[3] + fib[4];  
    ⋮           ⋮           ⋮
```

```
fib[0] = 1;  
fib[1] = 1;  
for( int i = 2; i < num; i++ )  
    fib[i] = fib[i-2] + fib[i-1];
```

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1							

```
fib[ 0 ] = 1;  
fib[ 1 ] = 1;  
for( int i{ 2 }; i < number; i++ )  
    fib[ i ] = fib[ i - 2 ] + fib[ i - 1 ];
```

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1	2						

```
fib[ 0 ] = 1;  
fib[ 1 ] = 1;  
for( int i{ 2 }; i < number; i++ )  
    fib[ i ] = fib[ i - 2 ] + fib[ i - 1 ];
```

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1	2	3					

```
fib[ 0 ] = 1;  
fib[ 1 ] = 1;  
for( int i{ 2 }; i < number; i++ )  
    fib[ i ] = fib[ i - 2 ] + fib[ i - 1 ];
```

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1	2	3	5				

```
fib[ 0 ] = 1;  
fib[ 1 ] = 1;  
for( int i{ 2 }; i < number; i++ )  
    fib[ i ] = fib[ i - 2 ] + fib[ i - 1 ];
```

Fibonacci numbers

	0	1	2	3	4	5	6	7	8
fib	1	1	2	3	5	8			

```
fib[ 0 ] = 1;  
fib[ 1 ] = 1;  
for( int i{ 2 }; i < number; i++ )  
    fib[ i ] = fib[ i - 2 ] + fib[ i - 1 ];
```

```
int main()
{
    int number;
    int fib[ 1000 ];
    cout << "Enter a positive integer ( 2 - 46 ) : ";
    cin >> number;
    fib[ 0 ] = 1;
    fib[ 1 ] = 1;
    for( int i{ 2 }; i < number; i++ )
        fib[ i ] = fib[ i - 2 ] + fib[ i - 1 ];

    for( int i{ 0 }; i < number; i++ )
        cout << fib[ i ] << endl;
    cout << endl;
}
```

Don't use arrays

Fibonacci numbers

```
int n1{ 1 };  
int n2{ 1 };  
int n3{ 2 };  
cout << n1 << endl;  
cout << n2 << endl;  
cout << n3 << endl;
```

num	46	0012FF18
		0012FF19
		0012FF1A
		0012FF1B
n1	1	0012FF1C
		0012FF1D
		0012FF1E
		0012FF1F
n2	1	0012FF20
		0012FF21
		0012FF22
		0012FF23
n3	2	0012FF2C
		0012FF2D
		0012FF2E
		0012FF2F

Fibonacci numbers

```
int n1{ 1 };  
int n2{ 1 };  
int n3{ 2 };  
cout << n1 << endl;  
cout << n2 << endl;  
cout << n3 << endl;
```

```
n1 = n2;  
n2 = n3;  
n3 = n1 + n2;  
cout << n3 << endl;
```

num	46	0012FF18
		0012FF19
		0012FF1A
		0012FF1B
n1	1	0012FF1C
		0012FF1D
		0012FF1E
		0012FF1F
n2	1	0012FF20
		0012FF21
		0012FF22
		0012FF23
n3	2	0012FF2C
		0012FF2D
		0012FF2E
		0012FF2F

Fibonacci numbers

```
int n1{ 1 };  
int n2{ 1 };  
int n3{ 2 };  
cout << n1 << endl;  
cout << n2 << endl;  
cout << n3 << endl;
```

```
n1 = n2;  
n2 = n3;  
n3 = n1 + n2;  
cout << n3 << endl;
```

num	46	0012FF18
		0012FF19
		0012FF1A
		0012FF1B
n1	1	0012FF1C
		0012FF1D
		0012FF1E
		0012FF1F
n2	2	0012FF20
		0012FF21
		0012FF22
		0012FF23
n3	2	0012FF2C
		0012FF2D
		0012FF2E
		0012FF2F

Fibonacci numbers

```
int n1{ 1 };  
int n2{ 1 };  
int n3{ 2 };  
cout << n1 << endl;  
cout << n2 << endl;  
cout << n3 << endl;  
  
n1 = n2;  
n2 = n3;  
n3 = n1 + n2;  
cout << n3 << endl;
```

num	46	0012FF18
		0012FF19
		0012FF1A
		0012FF1B
n1	1	0012FF1C
		0012FF1D
		0012FF1E
		0012FF1F
n2	2	0012FF20
		0012FF21
		0012FF22
		0012FF23
n3	3	0012FF2C
		0012FF2D
		0012FF2E
		0012FF2F

Fibonacci numbers

```
int n1{ 1 };  
int n2{ 1 };  
int n3{ 2 };  
cout << n1 << endl;  
cout << n2 << endl;  
cout << n3 << endl;
```

```
n1 = n2;  
n2 = n3;  
n3 = n1 + n2;  
cout << n3 << endl;
```

```
n1 = n2;  
n2 = n3;  
n3 = n1 + n2;  
cout << n3 << endl;
```

num	46	0012FF18
		0012FF19
		0012FF1A
		0012FF1B
n1	1	0012FF1C
		0012FF1D
		0012FF1E
		0012FF1F
n2	2	0012FF20
		0012FF21
		0012FF22
		0012FF23
n3	3	0012FF2C
		0012FF2D
		0012FF2E
		0012FF2F

Fibonacci numbers

```
int n1{ 1 };  
int n2{ 1 };  
int n3{ 2 };  
cout << n1 << endl;  
cout << n2 << endl;  
cout << n3 << endl;
```

```
n1 = n2;  
n2 = n3;  
n3 = n1 + n2;  
cout << n3 << endl;
```

```
n1 = n2;  
n2 = n3;  
n3 = n1 + n2;  
cout << n3 << endl;
```

num	46	0012FF18
		0012FF19
		0012FF1A
		0012FF1B
n1	2	0012FF1C
		0012FF1D
		0012FF1E
		0012FF1F
n2	2	0012FF20
		0012FF21
		0012FF22
		0012FF23
n3	3	0012FF2C
		0012FF2D
		0012FF2E
		0012FF2F

Fibonacci numbers

```
int n1{ 1 };  
int n2{ 1 };  
int n3{ 2 };  
cout << n1 << endl;  
cout << n2 << endl;  
cout << n3 << endl;
```

```
n1 = n2;  
n2 = n3;  
n3 = n1 + n2;  
cout << n3 << endl;
```

```
n1 = n2;  
n2 = n3;  
n3 = n1 + n2;  
cout << n3 << endl;
```

num	46	0012FF18
		0012FF19
		0012FF1A
		0012FF1B
n1	2	0012FF1C
		0012FF1D
		0012FF1E
		0012FF1F
n2	3	0012FF20
		0012FF21
		0012FF22
		0012FF23
n3	3	0012FF2C
		0012FF2D
		0012FF2E
		0012FF2F

Fibonacci numbers

```
int n1{ 1 };  
int n2{ 1 };  
int n3{ 2 };  
cout << n1 << endl;  
cout << n2 << endl;  
cout << n3 << endl;
```

```
n1 = n2;  
n2 = n3;  
n3 = n1 + n2;  
cout << n3 << endl;
```

```
n1 = n2;  
n2 = n3;  
n3 = n1 + n2;  
cout << n3 << endl;
```

num	46	0012FF18
		0012FF19
		0012FF1A
		0012FF1B
n1	2	0012FF1C
		0012FF1D
		0012FF1E
		0012FF1F
n2	3	0012FF20
		0012FF21
		0012FF22
		0012FF23
n3	5	0012FF2C
		0012FF2D
		0012FF2E
		0012FF2F

Fibonacci numbers

```
int number;  
int n1{ 1 };  
int n2{ 1 };  
int n3{ 2 };  
cin >> number;  
cout << n1 << endl;  
cout << n2 << endl;  
cout << n3 << endl;  
  
for( int i{ 4 }; i <= number; ++i )  
{  
    n1 = n2;  
    n2 = n3;  
    n3 = n1 + n2;  
    cout << n3 << endl;  
}
```

i	4	0012FF18
		0012FF19
		0012FF1A
		0012FF1B
n1	1	0012FF1C
		0012FF1D
		0012FF1E
		0012FF1F
n2	1	0012FF20
		0012FF21
		0012FF22
		0012FF23
n3	2	0012FF2C
		0012FF2D
		0012FF2E
		0012FF2F

Fibonacci numbers

```
int number;
int n1{ 1 };
int n2{ 1 };
int n3{ 2 };
cin >> number;
cout << n1 << endl;
cout << n2 << endl;
cout << n3 << endl;

for( int i{ 4 }; i <= number; ++i )
{
    n1 = n2;
    n2 = n3;
    n3 = n1 + n2;
    cout << n3 << endl;
}
```

i	4	0012FF18
		0012FF19
		0012FF1A
		0012FF1B
n1	1	0012FF1C
		0012FF1D
		0012FF1E
		0012FF1F
n2	2	0012FF20
		0012FF21
		0012FF22
		0012FF23
n3	2	0012FF2C
		0012FF2D
		0012FF2E
		0012FF2F

Fibonacci numbers

```
int number;  
int n1{ 1 };  
int n2{ 1 };  
int n3{ 2 };  
cin >> number;  
cout << n1 << endl;  
cout << n2 << endl;  
cout << n3 << endl;  
  
for( int i{ 4 }; i <= number; ++i )  
{  
    n1 = n2;  
    n2 = n3;  
    n3 = n1 + n2;  
    cout << n3 << endl;  
}
```

i	4	0012FF18
		0012FF19
		0012FF1A
		0012FF1B
n1	1	0012FF1C
		0012FF1D
		0012FF1E
		0012FF1F
n2	2	0012FF20
		0012FF21
		0012FF22
		0012FF23
n3	3	0012FF2C
		0012FF2D
		0012FF2E
		0012FF2F

Fibonacci numbers

```
int number;  
int n1{ 1 };  
int n2{ 1 };  
int n3{ 2 };  
cin >> number;  
cout << n1 << endl;  
cout << n2 << endl;  
cout << n3 << endl;  
  
for( int i{ 4 }; i <= number; ++i )  
{  
    n1 = n2;  
    n2 = n3;  
    n3 = n1 + n2;  
    cout << n3 << endl;  
}
```

i	5	0012FF18
		0012FF19
		0012FF1A
		0012FF1B
n1	1	0012FF1C
		0012FF1D
		0012FF1E
		0012FF1F
n2	2	0012FF20
		0012FF21
		0012FF22
		0012FF23
n3	3	0012FF2C
		0012FF2D
		0012FF2E
		0012FF2F

Fibonacci numbers

```
int number;  
int n1{ 1 };  
int n2{ 1 };  
int n3{ 2 };  
cin >> number;  
cout << n1 << endl;  
cout << n2 << endl;  
cout << n3 << endl;  
  
for( int i{ 4 }; i <= number; ++i )  
{  
    n1 = n2;  
    n2 = n3;  
    n3 = n1 + n2;  
    cout << n3 << endl;  
}
```

i	5	0012FF18
		0012FF19
		0012FF1A
		0012FF1B
n1	2	0012FF1C
		0012FF1D
		0012FF1E
		0012FF1F
n2	2	0012FF20
		0012FF21
		0012FF22
		0012FF23
n3	3	0012FF2C
		0012FF2D
		0012FF2E
		0012FF2F

Fibonacci numbers

```
int number;  
int n1{ 1 };  
int n2{ 1 };  
int n3{ 2 };  
cin >> number;  
cout << n1 << endl;  
cout << n2 << endl;  
cout << n3 << endl;  
  
for( int i{ 4 }; i <= number; ++i )  
{  
    n1 = n2;  
    n2 = n3;  
    n3 = n1 + n2;  
    cout << n3 << endl;  
}
```

i	5	0012FF18
		0012FF19
		0012FF1A
		0012FF1B
n1	2	0012FF1C
		0012FF1D
		0012FF1E
		0012FF1F
n2	3	0012FF20
		0012FF21
		0012FF22
		0012FF23
n3	3	0012FF2C
		0012FF2D
		0012FF2E
		0012FF2F

Fibonacci numbers

```
int number;  
int n1{ 1 };  
int n2{ 1 };  
int n3{ 2 };  
cin >> number;  
cout << n1 << endl;  
cout << n2 << endl;  
cout << n3 << endl;  
  
for( int i{ 4 }; i <= number; ++i )  
{  
    n1 = n2;  
    n2 = n3;  
    n3 = n1 + n2;  
    cout << n3 << endl;  
}
```

i	5	0012FF18
		0012FF19
		0012FF1A
		0012FF1B
n1	2	0012FF1C
		0012FF1D
		0012FF1E
		0012FF1F
n2	3	0012FF20
		0012FF21
		0012FF22
		0012FF23
n3	5	0012FF2C
		0012FF2D
		0012FF2E
		0012FF2F

```
int main()
{
    int number;
    int n1{ 1 };
    int n2{ 1 };
    int n3{ 2 };

    cout << "Enter a positive number ( 1 - 46 ) : ";
    cin >> number;
    cout << endl;

    cout << n1 << endl;
    cout << n2 << endl;
    cout << n3 << endl;

    for( int i{ 4 }; i <= number; ++i )
    {
        n1 = n2;
        n2 = n3;
        n3 = n1 + n2;
        cout << n3 << endl;
    }
}
```